

Shivani Chaudhary

B.Tech. Graduate (2023)

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Work Experience

Software Engineer | Balbix, Cybersecurity Company

(May'23- Present)

- **Fourth-Party Pipeline, API Optimization & AI Insights**
 - Built a **fourth-party discovery pipeline** by extending a distributed **ECS-based unified engine**, integrated with **Airflow and Snowflake** for scalable risk data enrichment across services
 - Engineered high-performance **Snowflake-backed APIs**, improving query latency and eliminating data inconsistencies, enabling **AI-driven summaries** across analytics and user-facing systems at scale
- **Agent Workflow & Event-Driven Systems**
 - Integrated configurable **third-party agents** using custom triggers and step types, supporting both **event-driven and manual workflow execution** of task and execution summaries
 - Built **bi-directional sync between tier settings and predefined workflow**, maintaining a single source of truth for configuration and ensuring consistent UI behavior and execution logic
- **CVE Management Dashboard**
 - Built a comprehensive **CVE management dashboard** that categorized vulnerabilities by asset, severity, exploitability, and fix availability, with intuitive **bar** and **donut** visualizations to improve vulnerability visibility
 - Added **advanced filtering and sorting** capabilities that streamlined triage workflows and **reduced time spent locating vulnerabilities by 40%**
- **API Development**
 - Developed Python **Flask** APIs serving **AI/ML** model data for vulnerability risk scoring and asset behavior analysis, handling **30–50 RPS** with reliable performance
 - Built **100+ GraphQL** queries using **Apollo Client, Node.js, and Express.js** with a **structured API framework** and robust **error handling**, cutting **UI disruptions by 30%**
- **Data Ingestion & Integration**
 - Engineered **connectors** to ingest asset data from Tenable, Qualys, CrowdStrike, Microsoft Defender, and ServiceNow CMDB, processing **5–15 GB** of data per customer weekly
 - **Sanitized** data from cybersecurity tools using schema validation, type normalization, deduplication, and JSON standardization, improving data quality for **20–30%**
 - Optimized data ingestion workflows, achieving a **15% efficiency increase** and enhancing the data pipeline's speed
- **Automated Data Export Solutions**
 - Automated weekly data export pipelines via Python scripts, extracting data from **Elasticsearch** and delivering JSON outputs to **Amazon S3** and via **Postman API**, reducing manual effort by **90%**
 - Implemented secure, scheduled API-based transfers using **Postman** and **cron**, ensuring reliable customer data delivery with **99%** data accuracy and zero manual intervention

Key Projects

- **AI-Powered Knowledge Base Search & Enrichment** 📄 (GitHub) (Self Project) (Oct'25)
 - Implemented an end-to-end **RAG pipeline** with **ingestion, chunking, SentenceTransformer embeddings, FAISS indexing**, SQLite storage, retrieval logic, and incremental updates
 - Built FastAPI-based semantic search and Q&A using Gemini, integrated with a Streamlit UI
- **Movie Recommendation System** 📄 (GitHub) (Self Project) (July'22)
 - Implemented backend part of Movie Recommendation and algorithms by using **Spyder** and **Jupyter Notebook**
 - Developed and optimized a **hybrid recommendation system** in **Python** using **machine learning** algorithms such as **KNN**, combining **content-based** and **collaborative filtering** to improve accuracy and relevance

Technical Skills

- **Programming Languages:** C, C++, Python, Golang, Java, JavaScript, TypeScript, SQL
- **Technologies & Tools:** React.js, Next.js, Node.js, Express.js, Kafka, Docker, Kubernetes, Git, Gitlab, LaTeX, MySQL, PostgreSQL, Elasticsearch, HTML5, CSS, Tailwind, Material-UI, Vite, GraphQL, Postman, phpMyAdmin, n8n, Jenkins, CI/CD Pipeline, AWS, SNS, SQS, Lambda, Airflow, Snowflake, RabbitMQ, Cursor, Jest, Redis, LLMs, RAG Pipeline, MCP server, DynamoDb, Distributed System